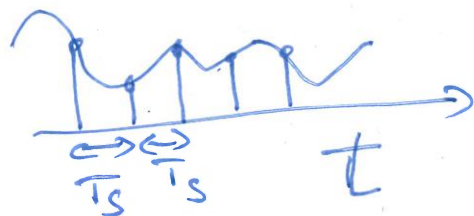


Analysis of Practical Observations

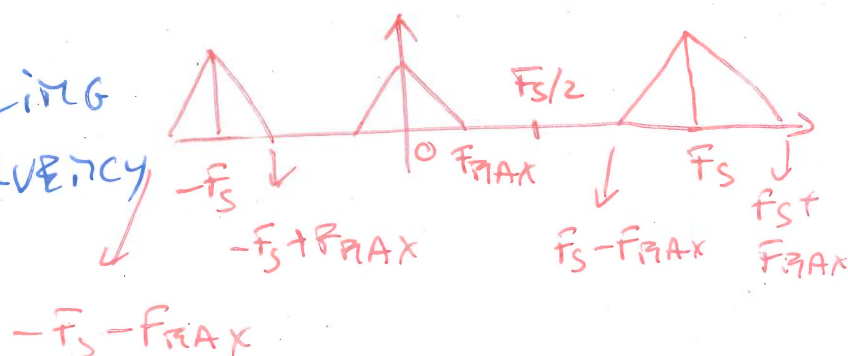
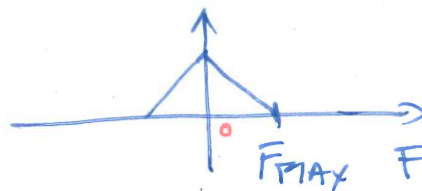
Time



$$F_s = \frac{1}{T_s}$$

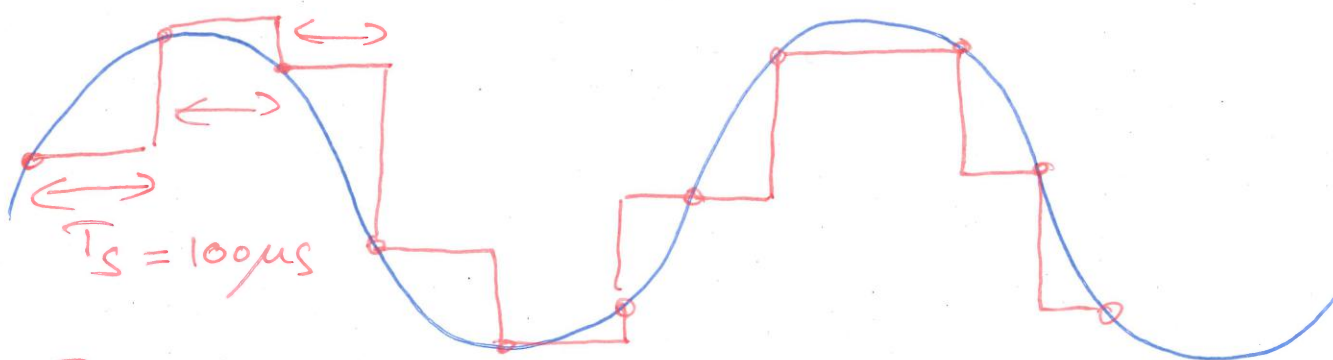
SAMPLING
FREQUENCY

Frequency



$$F_{MAX} < \frac{F_s}{2} \Rightarrow \underline{F_s > 2F_{MAX}}$$

$$F = 1 \text{ kHz}$$



$$F_s = \frac{1}{T_s} = 10 \text{ kHz}$$

$$= \frac{1}{100 \times 10^{-6}} \text{ Hz}$$

$$= \frac{10^6}{100} \text{ Hz}$$

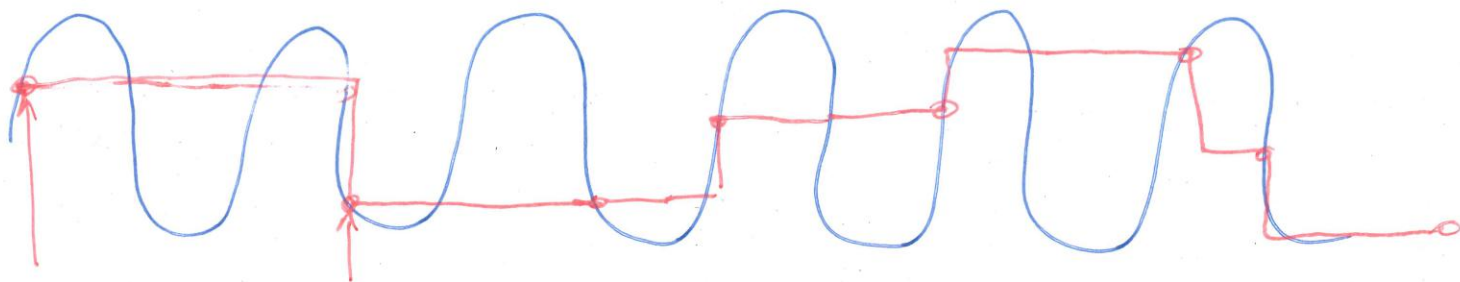
$$= 10^4 \text{ Hz}$$

$$= 10 \times 10^3 \text{ Hz}$$

$$= 10 \text{ kHz}$$

$$F' = F_S - F = 10 \text{ kHz} - 1 \text{ kHz} = 9 \text{ kHz}$$

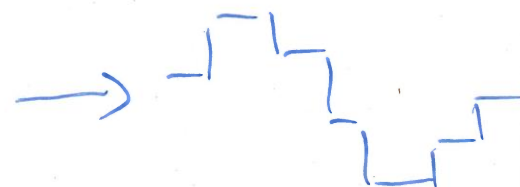
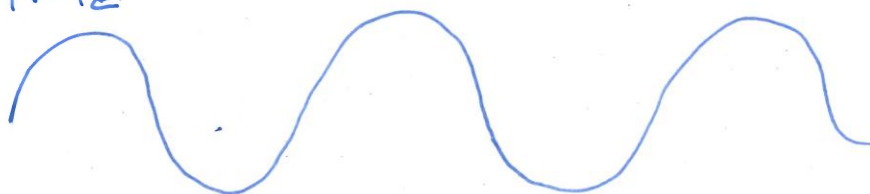
2



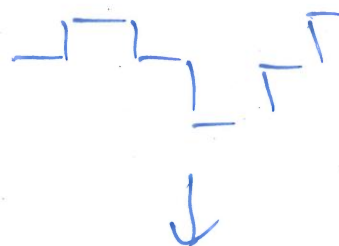
$$F'' = F_S + F = 10 \text{ kHz} + 1 \text{ kHz} = 11 \text{ kHz}$$

When does aliasing cause problems?

1 kHz

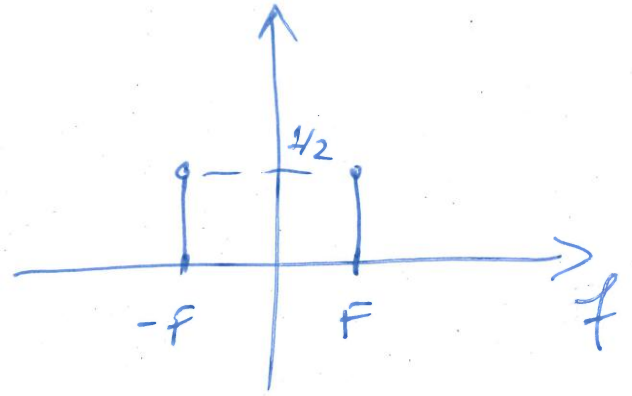
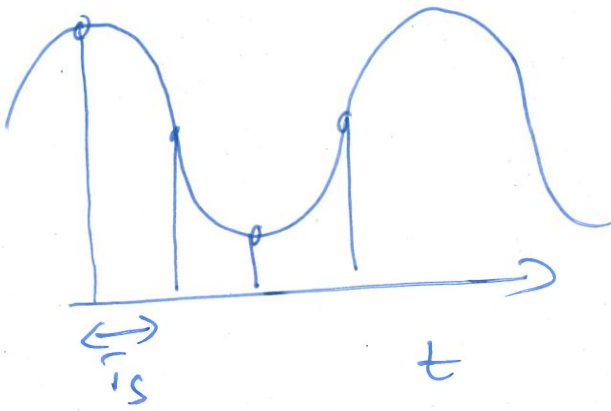


9 kHz

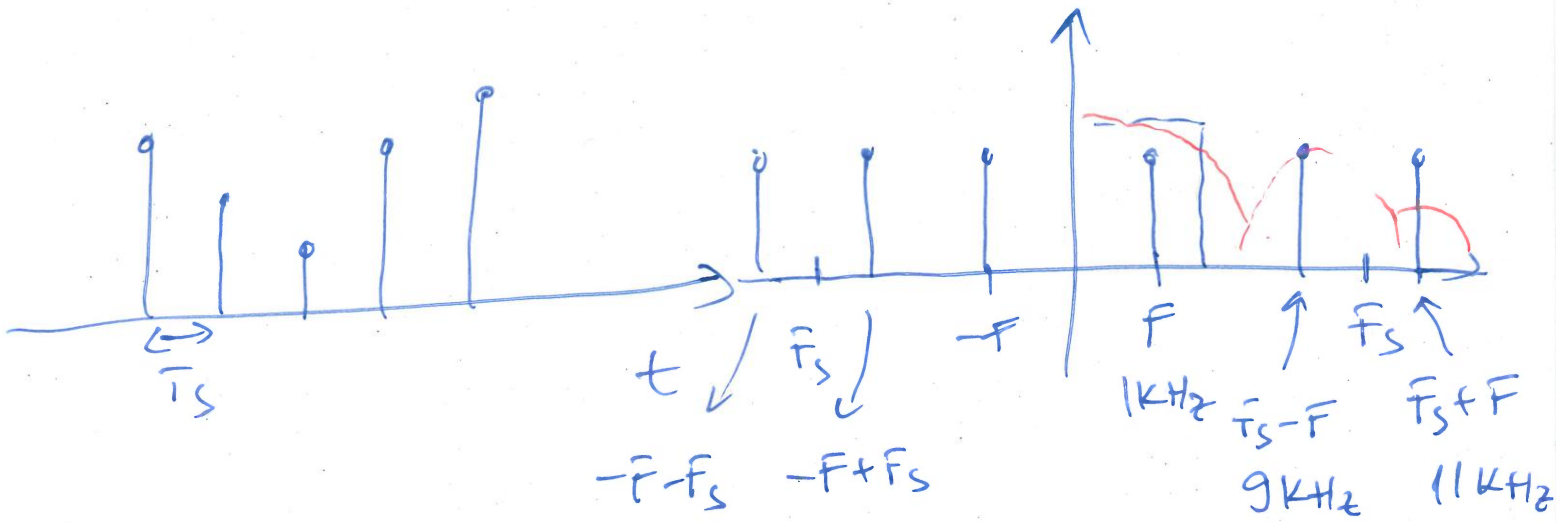


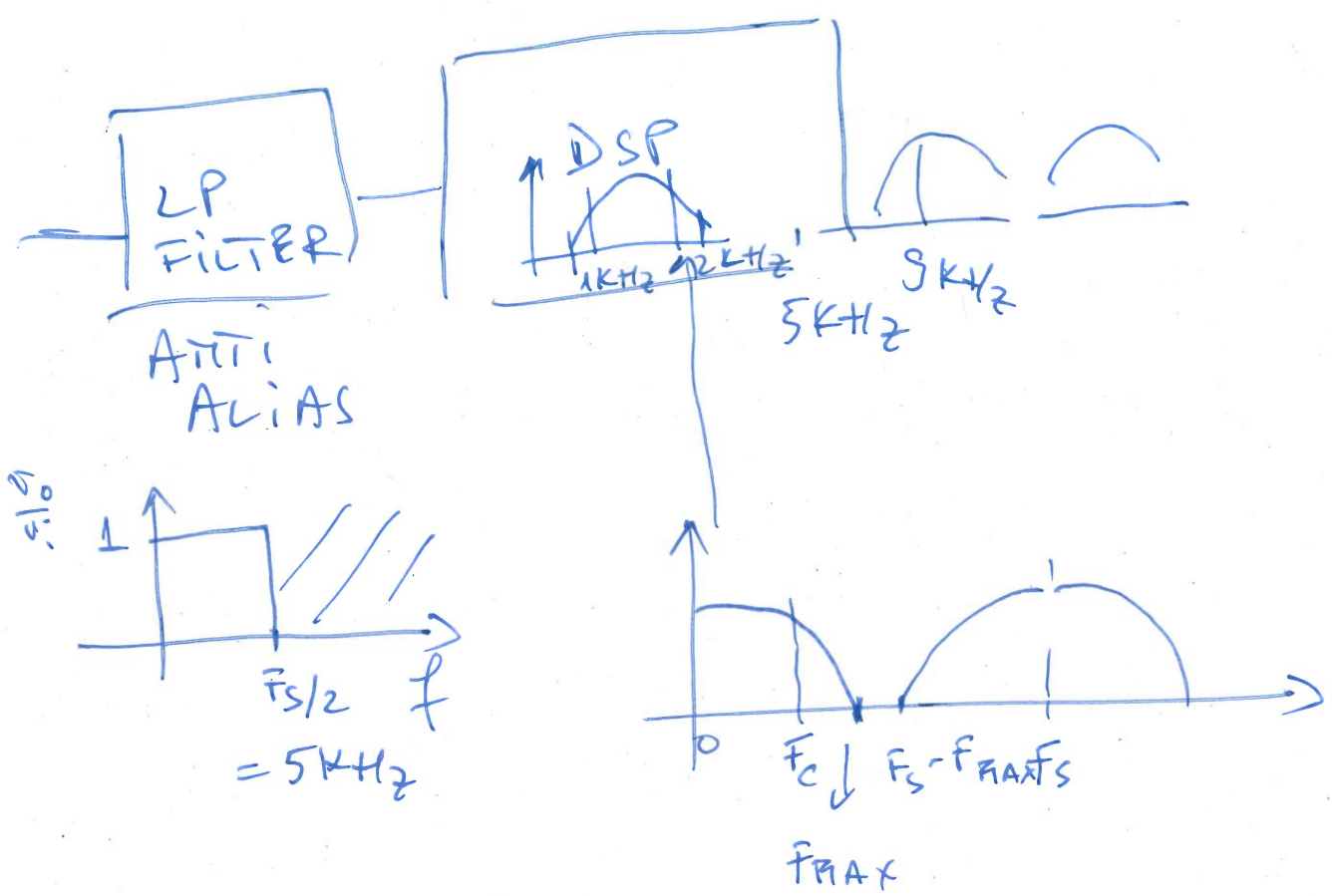
1 kHz

3



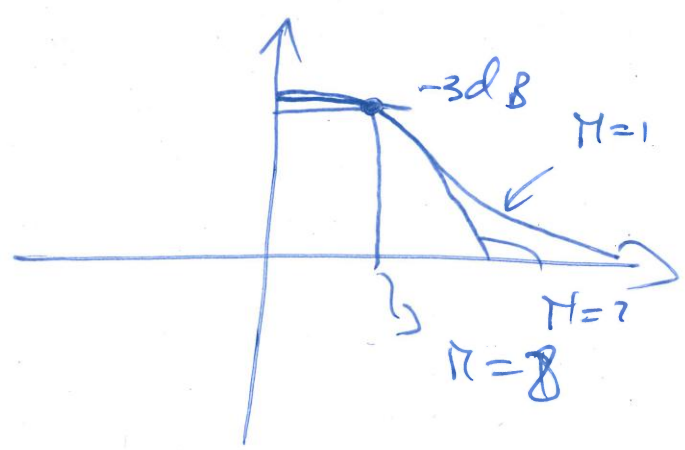
$$F_s = \frac{1}{T_s} = 10 \text{ kHz}$$





BUTTERWORTH

$$\frac{V_o}{V_i} = \frac{1}{\sqrt{1 + \left(\frac{f}{f_c}\right)^{2N}}}$$



$N=1$ BUTTERWORTH
RC

$N=2$ SALLER KEY

$$f_c = \frac{1}{2\pi RC}$$